AP BIOLOGY TOPIC 5.3

Mendelian Genetics, Probability Rules, Pedigrees & Chi-Square

UNIT 5: HEREDITY (8–11% EXAM WEIGHT)

EVO-2.A | IST-1.I | 3:1 & 9:3:3:1 | CHI-SQUARE (X^2)

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 5.3 (MENDELIAN GENETICS, PROBABILITY RULES, PEDIGREES & CHI-SQUARE)

CED Core Vocabulary

1. Law of Segregation — The two alleles for a heritable character separate (segregate) during gamete formation so that each gamete receives only 1 allele for each gene.

2. Law of Independent Assortment — Alleles of two (or more) different genes assort independently into gametes; inheriting one allele does not affect inheritance of another gene.

Genotype vs. Phenotype — Genotype = actual allele combination (TT , Tt , tt); Phenotype = observable physical/biochemical trait (TT and Tt share the same dominant phenotype; only tt...

Symbols — Squares = Males; Circles = Females; Shaded = Affected; Unshaded = Unaffected.

Meiotic Basis — Physical separation of homologous chromosomes during Anaphase I (and sister chromatids in Anaphase II).

Fertilization Restores Ploidy — Meiosis halves chromosome number (2n → n); fertilization fuses two haploid gametes (n + n → 2n) to restore the diploid state and create novel allele...

Homozygous vs. Heterozygous — Homozygous = identical alleles (AA dominant or aa recessive); Heterozygous = two different alleles (Aa).

Step 1 — Does the trait skip generations? — Two unaffected parents have an affected child → Trait MUST be Recessive (parents are heterozygous carriers, Aa × Aa → aa).

§2. MENDEL'S TWO LAWS & MEIOTIC CHROMOSOME BASIS

Chromosomal Basis

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Mendel's Two Laws & Meiotic Chromosome Basis ➔ **1. Law of Segregation** ➔ **2. Law of Independent Assortment** ➔ **Fertilization Restores Ploidy**

- **1. Law of Segregation:**
  - The two alleles for a heritable character separate (segregate) during gamete formation so that **each gamete receives only 1 allele** for each gene.
  - **Meiotic Basis:** Physical separation of homologous chromosomes during **Anaphase I** (and sister chromatids in Anaphase II).
- **2. Law of Independent Assortment:**
  - Alleles of two (or more) different genes assort independently into gametes; inheriting one allele does not affect inheritance of another gene.
  - **Meiotic Basis:** Random alignment of homologous chromosome pairs along the metaphase plate during **Metaphase I**.
- **Fertilization Restores Ploidy:** Meiosis halves chromosome number (**2n → n**); fertilization fuses two haploid gametes (**n + n → 2n**) to restore the diploid state and create novel allele combinations in the zygote.

[AP EXAM FRQ SCORING STRATEGY] **When Independent Assortment Applies:** Genes assort independently when located on **different chromosomes** (unlinked) or very far apart (**RF = 50%**) on the same chromosome.

§5. PEDIGREE ANALYSIS: RAPID DIAGNOSTIC FLOWCHART

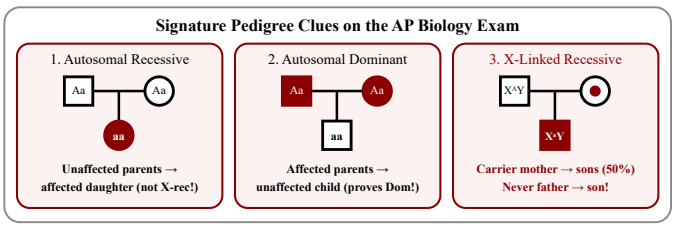
Autosomal vs. Sex-Linked

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Pedigree Analysis ➔ **Symbols** ➔ **Step 1 — Does the trait skip generations?**

- **Symbols:** Squares = Males; Circles = Females; Shaded = Affected; Unshaded = Unaffected.
- **Step 1 — Does the trait skip generations?**
  - **Two unaffected parents have an affected child** → Trait MUST be **Recessive** (parents are heterozygous carriers, **Aa × Aa → aa**).
  - **Two affected parents have an unaffected child** → Trait MUST be **Dominant** (parents are heterozygotes, **Aa × Aa → aa**).
- **Step 2 — Check Sex Bias & Father-to-Son Transmission:**
  - **Autosomal Recessive:** Affects males & females equally; skips generations (e.g., Cystic fibrosis, Tay-Sachs, PKU).
  - **Autosomal Dominant:** Affects males & females equally; appears in **every generation**; affected child always has ≥1 affected parent (e.g., Huntington's, Achondroplasia).
  - **X-Linked Recessive:** Mainly affects **males (X^aY)**; **NO father-to-son transmission** (fathers give Y to sons); affected female (X^aX^a) *must* have an affected father and a carrier/affected mother.
  - **X-Linked Dominant:** Affected father (X^AY) passes trait to **100% of his daughters** and **0% of his sons**.

Signature Pedigree Clues on the AP Biology Exam



§3. GENETIC VOCABULARY & STANDARD MENDELIAN CROSSES

3:1, 9:3:3:1, Testcross

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Genetic Vocabulary & Standard Mendelian Crosses ➔ **Genotype vs. Phenotype** ➔ **Homozygous vs. Heterozygous**

- **Genotype vs. Phenotype:** **Genotype** = actual allele combination (**TT**, **Tt**, **tt**); **Phenotype** = observable physical/biochemical trait (**TT** and **Tt** share the same dominant phenotype; only **tt** shows the recessive phenotype).
- **Homozygous vs. Heterozygous:** **Homozygous** = identical alleles (**AA** dominant or **aa** recessive); **Heterozygous** = two different alleles (**Aa**).

CROSS TYPE	PARENTAL CROSS	GENOTYPIC & PHENOTYPIC RATIOS
Monohybrid Cross (1 gene)	**Aa × Aa**	**Genotype: 1 AA : 2 Aa : 1 aa** **Phenotype:** **3 Dominant : 1 Recessive**
Dihybrid Cross (2 unlinked genes)	**AaBb × AaBb**	**Phenotype:** **9 A_B_ : 3 A_bb : 3 aaB_ : 1 aabb** *(Two simultaneous 3:1 crosses: (3:1)×(3:1) = 9:3:3:1)*
Test Cross (unknown dominant **A_**)	**A_ × aa** (homozygous rec.)	If **AA × aa** → **100% dominant (Aa)** If **Aa × aa** → **1:1 ratio (50% Aa : 50% aa)**
Dihybrid Test Cross	**AaBb × aabb**	**1 : 1 : 1 : 1 (25% each: AaBb, Aabb, aaBb, aabb)** if unlinked

[COMMON AP EXAM FRQ TRAP] **Dominant ≠ Most Common in a Population:** Dominant only means the allele masks the recessive allele in a heterozygote (**Aa**). A dominant allele (like Huntington's disease or polydactyly) can be rare in a population!

§4. RULES OF PROBABILITY IN MULTI-GENE CROSSES

Skip Giant Punnett Squares!

- **Multiplication Rule ("AND" Rule — Independent Events):**
  - Probability that two or more independent events occur together equals the **product** of their individual probabilities: **P(A and B) = P(A) × P(B)**.
- **Addition Rule ("OR" Rule — Mutually Exclusive Events):**
  - Probability that an event can occur in two or more mutually exclusive ways equals the **sum** of their individual probabilities: **P(A or B) = P(A) + P(B)**.

[EXAMPLE & BIOLOGICAL APPLICATION] **Fast Trihybrid Cross (AaBbCc × AaBbcc):** What is the probability of an **aabbcc** offspring? Treat each gene as a separate monohybrid cross and multiply: **P(aa) = ¼**, **P(bb) = ¼**, **P(cc) = ½** → **¼ × ¼ × ½ = 1/32**!

§6. CHI-SQUARE (X^2) GOODNESS-OF-FIT TEST IN GENETICS

AP Formula Sheet

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Chi-Square (χ^2) Goodness-of-Fit Test in Genetics ➔ **Purpose** ➔ **Null Hypothesis (H_0)** ➔ **Degrees of Freedom (df)**

$\chi^2 = \Sigma [(\text{Observed} - \text{Expected})^2 / \text{Expected}]$ | Degrees of Freedom: df = n − 1

- **Purpose:** Tests whether deviations between **Observed (O)** offspring counts and **Expected (E)** Mendelian counts are due to random chance or a different inheritance pattern (e.g., linkage).
- **Null Hypothesis (H_0):** There is **no statistically significant difference** between observed and expected counts (the data follow the predicted Mendelian ratio; any difference is due to chance).
- **Degrees of Freedom (df):** Number of phenotypic categories minus 1 (**df = n − 1**). Use critical value at **p = 0.05**:
  - If **χ^2 < Critical Value** → **Fail to reject H_0** (data fit Mendelian model; deviations are random chance).
  - If **$\chi^2 \geq$ Critical Value** → **Reject H_0** (data do NOT follow Mendelian model; e.g., genes are linked!).

[EXAMPLE & BIOLOGICAL APPLICATION] **Worked χ^2 Example:** In a dihybrid test cross (**AaBb × aabb**, N = 400 total offspring), expected 1:1:1:1 ratio gives **E = 100** for each of the 4 classes (**df = 4 − 1 = 3**, critical value at **p=0.05** is **7.82**). If observed counts are 170, 170, 30, 30, then **χ^2 = 49 + 49 + 49 + 49 = 196 > 7.82** → **Reject H_0** (genes are linked on the same chromosome!).

§7. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 5.3 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 5.3: Mendelian Genetics, Probability Rules, Pedigrees & Chi-Square ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigation 7 Part 2 (Sordaria fimicola Meiotic Crossing Over) & Drosophila/Fast Plants Crosses: Counting 4:4 non-recombinant vs. 2:2:2:2 / 2:4:2 recombinant asci in Sordaria perithecia to determine gene-to-centromere map distance (% Crossover = [Recombinant asci / Total asci] × 100; Map distance in cM = % Crossover / 2 because only 2 of 4 chromatids in a tetrad cross over). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Statistical Testing: Chi-Square Goodness-of-Fit: $\chi^2 = \Sigma [(O - E)^2 / E]$, with Degrees of Freedom df = (number of phenotypic classes − 1). At $\alpha = 0.05$: if $\chi^2 >$ critical value, REJECT H_0 (e.g., genes are autosomal linked rather than assorting independently 9:3:3:1); if $\chi^2 \leq$ critical value, FAIL TO REJECT H_0. Recombination Frequency: RF = (Total Recombinants / Total Offspring) × 100% = Map Units (cM). **Error Bar Rule (±2 SE_x̄):** If the ±2 $SE_{\bar{x}}$ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Chromosomal Disruption Case Study: Meiotic Nondisjunction (failure of homologous chromosomes to separate in Anaphase I, or sister chromatids to separate in Anaphase II) yields aneuploid gametes (n + 1 or n − 1). Fertilization produces Trisomy 21 (Down Syndrome, 47, +21), Klinefelter Syndrome (47, XXY), or monosomic Turner Syndrome (45, X0). Mitochondrial myopathies (e.g., Leber Hereditary Optic Neuropathy) exhibit strict maternal inheritance because the ovum contributes virtually all zygotic cytoplasm and mitochondria.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 5.3 to prove why the claim holds.

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